

# Virtual Impedance Current Limiting for Inverters in Microgrids With Synchronous Generators

Andrew D. Paquette, *Member, IEEE*, and Deepak M. Divan, *Fellow, IEEE*

**Abstract**—This paper focuses on current limiting for voltage-controlled inverters during overloads caused by poor transient load sharing between inverters and synchronous generators in islanded microgrids. The use of simple current reference saturation limiters can cause instability when the voltage regulator loses control after the current reference saturates. The use of virtual impedance for current limiting is shown to improve transient stability during current limiting when operating in parallel with synchronous generators. Small-signal analysis is used to set the virtual impedance magnitude and  $X/R$  ratio, and validation is provided by simulation and experimental results.

**Index Terms**—Current control, current limiting, generators, inverters, microgrids, virtual impedance, voltage control.

## I. INTRODUCTION

FOR STABLE islanded operation, a microgrid requires at least one source that is able to regulate voltage and frequency and respond quickly to changes in load. This requires some form of energy storage or a fast-responding energy source. For microgrids, the practical choices are generators, or inverters with energy storage or a fast-responding energy source. There are four basic types of inverter control [1]–[3], as shown in Fig. 1. Grid-forming control acts as a fixed voltage source, and grid-feeding control acts as a fixed current source. Grid-supporting-grid-forming (GSGFm) control acts as a droop-controlled voltage source, where voltage and frequency references are adjusted based on measured real and reactive power. Grid-supporting-grid-feeding control acts as a droop-controlled current source, where real and reactive power references are adjusted based on measured voltage and frequency. Grid-forming control is not suitable for paralleling with other voltage sources, and both grid-feeding and grid-supporting-grid-feeding controls require a voltage source to be online also [1], [2]. GSGFm control allows multiple grid-forming sources to operate in parallel in grid connected or islanded mode [1], [3]–[6], eliminating the need for rapid control-mode switching when generator(s) transition on and off, or when switching from grid-connected to islanded operation.

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The authors are with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA 30332 USA (e-mail: adpaquet@gatech.edu; ddivan@ece.gatech.edu).

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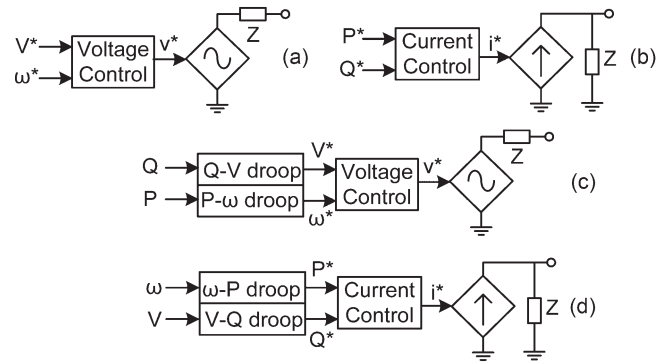
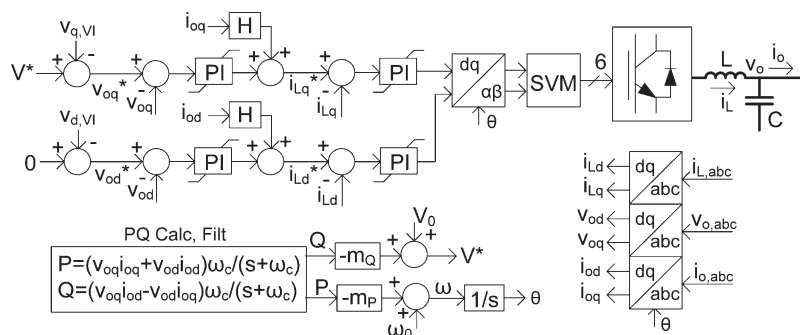


Fig. 1. Four basic types of inverter control. (a) Grid-forming, (b) grid-feeding, (c) GSGFm, and (d) grid-supporting-grid-feeding [3].

When synchronous generators and GSGFm inverters operate together in islanded mode, they exhibit poor transient load sharing, where the inverter picks up the majority of any load step [3], [7], [8]. This constrains the inverter rating to be able to handle the largest expected load step and could negatively impact the life of battery energy storage inverters. It is possible to modify the inverter controls to improve the transient load sharing, but this comes at the expense of increased voltage and frequency transients [3]. Instead of trying to improve the transient load sharing by effectively slowing down the inverter, it is desirable to fully utilize the inverter by allowing it to provide as much support as possible and simply limit current when necessary. This takes advantage of the fast response characteristics of the inverter to improve power quality without overloading the inverter during transients, as well as protecting the inverter during faults.

Current limiting is accomplished differently in single-loop and multiloop voltage control, the two main types of inverter voltage control. Current limiting in single-loop voltage control is generally accomplished by reducing the voltage magnitude when the current exceeds a threshold, through a predefined droop relationship [9], or by a current-limiting proportional–integral (PI) controller [10]. In multiloop voltage control methods, the two main current-limiting methods are to either limit the current [11]–[13] or to limit the voltage reference with a virtual impedance [14]–[17].

Virtual impedance has been proposed for many purposes, including improving damping, mitigating impact of unequal or resistive line impedances, and for current limiting [1], [6], [14]–[18]. An early example of virtual impedance for current limiting is [19], but for use in a series converter. Virtual resistance for current limiting is shown in [14] and [17], and a comprehensive design of virtual impedance, as well as the current-limiting



implementation from which this paper's virtual impedance current limiting is derived, is shown in [15].

Limiting inverter current magnitude is relatively simple, but maintaining stability and exiting current limiting is challenging, particularly when dissimilar voltage sources such as synchronous generators and GSGFm inverters are paralleled. Out of [9]–[17], only [9], [11], [12], [17] address how to exit current limiting, and [9], [11], [17] only consider a single inverter, for which entering and exiting current limiting is relatively straightforward. Reference [12] addresses current limiting with inverters and synchronous generators but uses a complicated control scheme. Additional discussion on [12] is provided in Section III-A.

This paper identifies that when GSGFm inverters go into current-limiting mode during overloads in parallel with synchronous generators, the use of simple current reference saturation limiters may cause instability. When the current reference saturates, the voltage controller effectively disabled may lose control. The voltage and current controllers can then “wind up” as the inverter and generator frequencies diverge, and the system becomes unstable. The use of virtual impedance current limiting is proposed to limit the current by limiting the voltage reference and preventing the voltage regulator from commanding an excessive current.

This paper examines the current-limiting characteristics of multiloop GSGFm inverter controls [1], [4]–[6], specifically looking at current limiting during overloads caused by poor transient load sharing with synchronous generators. Section II describes the multiloop  $dq$  GSGFm control, along with a proposed transient virtual impedance to eliminate steady-state virtual impedance voltage drop. In Section III, it is shown that the use of simple current reference saturation can cause instability. Section IV proposes virtual impedance current limiting to improve transient stability by avoiding saturation of the current reference. Finally, conclusions are given in Section V.

### A. Basics

Multiloop voltage control uses an outer voltage loop that provides the current reference to an inner current loop. In GSGFm controls, the voltage reference and frequency are given by voltage and frequency droop. The multiloop  $dq$  voltage control used in this paper is shown in Fig. 2. The voltage reference

and frequency are given by conventional voltage and frequency droop. Virtual output impedance is added by subtracting the voltage drop across a virtual resistance and inductance from the voltage reference. The virtual impedance voltage drop calculation is described in the next section. The outer voltage loop includes output current feedforward (with gain  $H$ ) [4], [5], [12]. Other feedforward and compensation terms were investigated in this paper (specifically voltage feedforward and decoupling terms in the current controller, and filter capacitor current compensation in the voltage controller [4], [5], [12]) but have been omitted because small-signal analysis indicated they had negligible impact on any of the important modes. In addition, in the presence of significant measurement noise in the experimental setup, some of the compensation terms significantly degraded performance by feeding forward noise into the current and voltage commands.

Virtual output impedance degrades voltage regulation due to steady-state voltage drop across the virtual impedance. It has been proposed to use a supervisory or tertiary control [1], [20] to adjust each source's voltage reference to compensate for virtual impedance voltage drops. It should be noted that supervisory or tertiary control can also be used for many other functions, e.g., maintaining nominal voltage and frequency throughout the microgrid, adjusting power set points, and turning sources on/off based on demand. Eliminating steady-state virtual impedance voltage drop is particularly important when operating in parallel with a synchronous generator, as a steady-state voltage drop causes significant reactive power sharing error because the generator's automatic voltage regulator (AVR) does not include a steady-state voltage drop. To avoid the need for a central controller to continuously modify all of the voltage references, a variation on the conventional virtual impedance is proposed.

This paper proposes using a transient virtual impedance, wherein a high-pass filtered version of the  $dq$  output current is used in the virtual impedance calculation. In the  $dq$  frame, the voltage drop across an RL impedance is given by (1) and (2)

$$v_d = Ri_d - \omega Li_q \quad (1)$$

$$v_g = Ri_g + \omega Li_d. \quad (2)$$

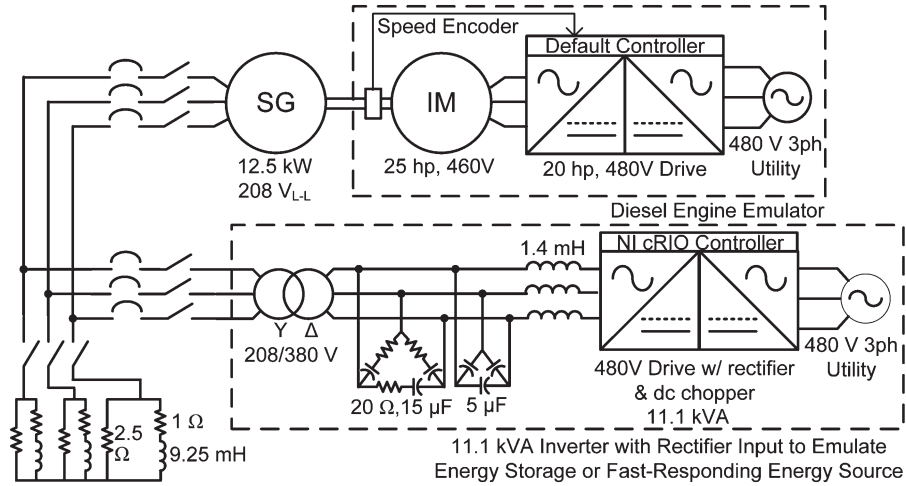


Fig. 3. Diagram of experimental microgrid setup.

The transient virtual impedance voltage drop is then given by (3) and (4), where  $\omega_{c,hpf}$  is the cutoff frequency of a single-order high-pass filter, and  $R_{VI}^0$  and  $L_{VI}^0$  are the virtual resistance and inductance, respectively,

$$v_{d,VI} = R_{VI}^0 i_{od} s / (s + \omega_{c,hpf}) - \omega L_{VI}^0 i_{oq} s / (s + \omega_{c,hpf}) \quad (3)$$

$$v_{q,VI} = R_{VI}^0 i_{oq} s / (s + \omega_{c,hpf}) + \omega L_{VI}^0 i_{od} s / (s + \omega_{c,hpf}). \quad (4)$$

The zero superscript is used to indicate a constant value, because a nonconstant virtual impedance term is added in Section IV for current limiting. The positive sequence fundamental component of the current is constant in the synchronous  $dq$  frame, and thus, the positive sequence fundamental virtual impedance voltage drop will decay to zero in steady state. This allows the virtual impedance to impart the necessary stabilizing effects without the steady-state voltage drop.

It should be noted that in this paper, the inverter uses an output transformer, and the voltage drop across the transformer is not currently being compensated. The extra output impedance from the transformer causes steady-state reactive power sharing error. If it is desired to compensate for the voltage drop across the output transformer, a slower, extra outer voltage loop could be used to modify the primary voltage loop's reference to slowly restore the transformer output voltage to the droop set point. In the case of an extra outer voltage loop, transient virtual impedance would not be necessary, since the outer loop would compensate for the steady-state voltage drop across both the virtual output impedance and the transformer impedance.

### III. CURRENT LIMITING IN MULTILoop INVERTER CONTROLS

#### A. Current Reference Saturation

The multiloop voltage control in Fig. 2 has inherent current-limiting ability due to the use of current regulators. Assuming the current regulators have sufficiently high bandwidth, the current can be limited simply by limiting the current reference, i.e.,  $i_{Ld}^*$  and  $i_{Lq}^*$ .

With current-controlled inverters, i.e., grid-feeding control, the phase-lock-loop (PLL) tracks the grid voltage phase angle

and aligns the  $dq$  transformation angle with the grid voltage, such that the  $d$ - and  $q$ -axis currents correspond to reactive and real current, respectively. The multiloop GSGFm strategy in Fig. 2 aligns the  $q$ -axis with the output voltage by setting the  $d$ -axis voltage reference to zero and the  $q$ -axis voltage reference to the desired voltage magnitude. In a grid-feeding inverter, the PLL maintains phase angle alignment even during current limiting (except for some transient error). However, with GSGFm control, the output of the voltage regulator(s) saturates once current limiting starts, and the voltage regulators are not necessarily able to keep the  $q$ -axis aligned with the output voltage (or grid voltage). While the current magnitude is effectively controlled, the alignment of the  $d$ - and  $q$ -axis is not maintained. This allows the phase angle to drift, and if another voltage source is present with a different frequency, such as a generator whose rotor speed is transiently dipping following a load step, the frequencies may diverge causing the  $dq$  voltages to “wind up.” This problem is illustrated through a simulation in Section III-B.

Reference [12] uses a hybrid between grid-forming and grid-feeding control and, thus, may not suffer from the instability aforementioned. In [12], the  $dq$  transformation angle is provided by a PLL as in grid-feeding control, but there are extra regulation loops that attempt to drive the frequency and phase angle to that given by the droop frequency (similar to GSGFm control). Since the PLL tracks the output voltage phase angle during current limiting, the inverter  $dq$  transformation angle should not drift from the output voltage angle. However, the control in [12] is very complex, with many extra regulation loops. The control proposed in this paper has the advantage of significantly reduced complexity.

#### B. Simulation

The system diagram for the experimental microgrid setup is shown in Fig. 3. The inverter has an output delta-wye transformer for galvanic isolation to enable supplying single-phase loads with a three-wire inverter. Controls for a generator with droop are shown in Fig. 4. The generator controls are standard PID controls for the governor and AVR, with droop

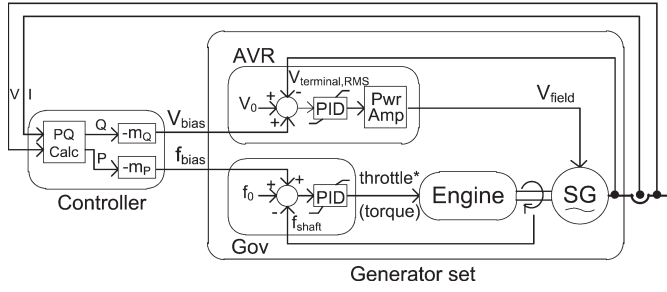


Fig. 4. Control diagram for synchronous generator with droop.

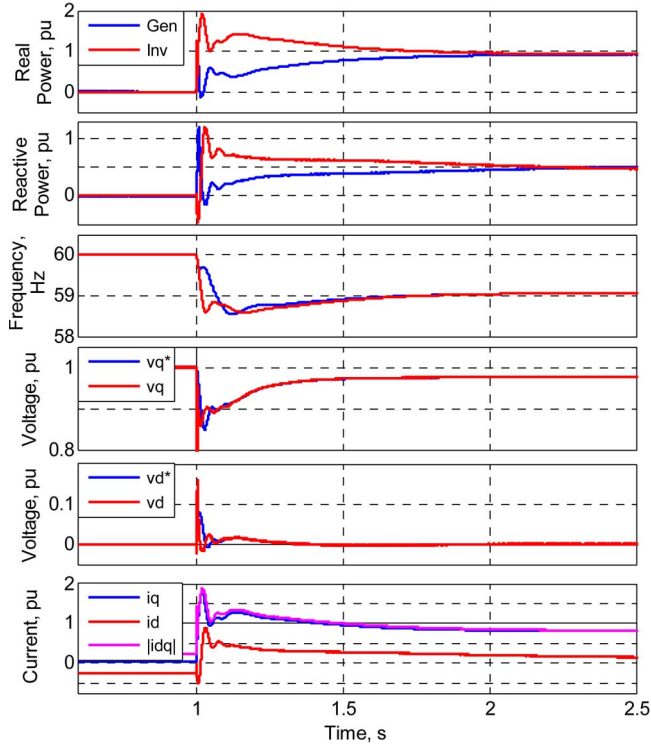


Fig. 5. Simulation of base case response to application of 21 kW 0.9 power factor load without current limiting.

implemented by biasing the voltage and frequency references. However, generator droop is implemented slightly differently in the experimental setup, with reactive droop implemented in the AVR [21] using line-line voltage and phase B current, and speed droop implemented in the induction motor drive [22] using estimated motor torque.

A simulation of the response of the baseline controls [see Fig. 2 and (3) and (4)] to the application of a 21 kW 0.9 power factor linear load is shown in Fig. 5. The inverter initially picks up most of the load step, and the current reaches 1.89 pu. In Fig. 6, the same load step is applied, except with the current reference saturation limiters set at  $\pm 1.06$  pu ( $I_{\text{mag}} = \sqrt{1.06^2 + 1.06^2} = 1.5$  pu). The  $q$ -axis current reference saturates first, and as the  $q$ -axis voltage drifts, the output voltage alignment drifts and the  $d$ - and  $q$ -axis currents no longer correspond to real and reactive power. As the current references drift, the load on the generator drifts. When the generator's real power output increases too far, the generator slows down excessively, and the generator rotor speed and inverter frequency

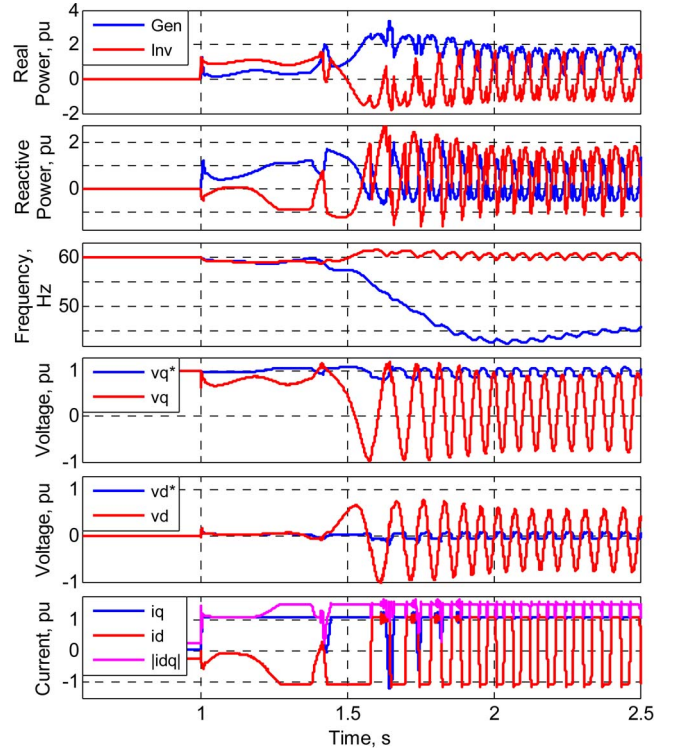


Fig. 6. Simulation of response to application of 21 kW 0.9 power factor load with current limiting, showing instability caused by current reference saturation limiters.

diverge. As the inverter and generator frequencies diverge, the inverter  $d$ - and  $q$ -axis voltage begin to “wind up” and oscillate at a frequency equal to the difference between the generator and inverter frequency. As the sign of the  $d$ - and  $q$ -axis voltage error changes, the current reference flip-flops between the positive and negative limits, and the voltage regulators never regain control. Note that, in this paper, the current references are not shown in the plots, since on the plotting timescale, the current regulator tracking error is negligible. In addition, generator and inverter frequencies in the plots are generator rotor speed and inverter droop frequency ( $\omega_0 - m_P * P$ ), respectively.

#### IV. VIRTUAL IMPEDANCE CURRENT LIMITING

Virtual impedance can be used to increase the inverter's output impedance during transients and thereby limit the current [14]–[16]. While normal current limiting works by limiting the current reference, virtual impedance current limiting works by reducing the voltage reference, thereby preventing the voltage controller from commanding an excessively large current reference. During current limiting, the voltage controller output does not saturate, and the instability observed in Fig. 6 is avoided.

When the current magnitude exceeds a threshold, additional virtual impedance is added. The total virtual impedance is given by (5) and consists of a nominal virtual impedance, i.e.,  $R_{VI}^0$  and  $L_{VI}^0$ , plus a current limiting virtual impedance, i.e.,  $\Delta R_{VI}$  and  $\Delta L_{VI}$ . This is the same as in [15]. The expressions for  $\Delta R_{VI}$  and  $\Delta L_{VI}$  are given in (6) and (7), where  $k_{p,Rvi}$ ,  $\Delta X/R$ , and  $I_{\text{thresh}}$  are the current limiting virtual resistance proportional gain, current limiting virtual impedance  $X/R$  ratio, and



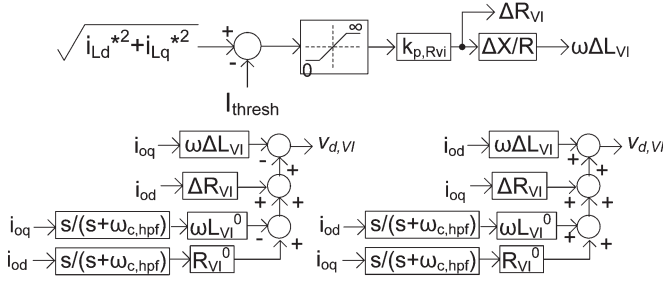


Fig. 7. Implementation of the virtual impedance calculation.

current limiting threshold, respectively.  $\Delta X/R$  is used to set the ratio  $\omega \Delta L_{VI}/\Delta R_{VI}$

$$R_{VI} = R_{VI}^0 + \Delta R_{VI}, L_{VI} = L_{VI}^0 + \Delta L_{VI} \quad (5)$$

$$\Delta R_{VI} = \max \left( k_{p,Rvi} \left( \sqrt{i_{Ld,ref}^2 + i_{Lq,ref}^2} - I_{thresh} \right), 0 \right) \quad (6)$$

$$\Delta L_{VI} = \Delta R_{VI}(\Delta X/R). \quad (7)$$

For current limiting, transient virtual impedance is undesirable because the virtual impedance voltage drop decays with the high-pass filter time constant. Therefore, in this paper, transient virtual impedance is used for  $Z_{VI}^0$ , but normal (not high-pass filtered) virtual impedance is used for  $\Delta Z_{VI}$ . The final expression used in this paper for the virtual impedance voltage drop is given by (8) and (9), with  $\Delta R_{VI}$  and  $\Delta L_{VI}$  given by (6) and (7),

$$v_{d,VI} = R_{VI}^0 i_{od} s / (s + \omega_{c,hpf}) - \omega L_{VI}^0 i_{oq} s / (s + \omega_{c,hpf}) + \Delta R_{VI} i_{od} - \Delta \omega L_{VI} i_{oq} \quad (8)$$

$$v_{q,VI} = R_{VI}^0 i_{oq} s / (s + \omega_{c,hpf}) + \omega L_{VI}^0 i_{od} s / (s + \omega_{c,hpf}) + \Delta R_{VI} i_{oq} + \Delta \omega L_{VI} i_{od}. \quad (9)$$

The virtual impedance implementation is shown in Fig. 7. Note that (6) and (7) use  $i_{Ldq,ref}$  for the current magnitude calculation instead of  $i_{odq}$  because using  $i_{odq}$  was found to cause significantly more oscillatory behavior.

It should be noted that the nonfiltered virtual impedance used for current limiting behaves similarly to the high-pass filtered virtual impedance during transients. The high-pass filter allows the transient current to pass but eliminates the steady-state virtual impedance voltage drop caused by the positive sequence fundamental current, which is dc in the synchronous  $dq$  frame. The current-limiting virtual impedance is not high-pass filtered, because if the virtual impedance voltage drop decayed, it would not limit the current. The overall virtual impedance and damping with and without the current-limiting virtual impedance will be significantly different. However, the difference is due primarily to the addition of a large virtual impedance rather than whether or not the current is high-pass filtered.

### A. Setting Current-Limiting Gains

The choice of the current-limiting gains  $k_{p,Rvi}$  and  $k_{p,Rvi} * \Delta X/R$  are chosen to limit the current magnitude to a suitable level during a bolted fault. For a three-phase bolted fault, the virtual impedance voltage drop should equal the voltage magnitude command, i.e.,  $V_0$ , as given by (10), where  $I_{max}$  is the desired maximum current magnitude

$$V_0 = I_{max} Z_{VI} = I_{max} \sqrt{R_{VI}^2 + (\omega L_{VI})^2}. \quad (10)$$

For given values of  $I_{max}$ ,  $I_{thresh}$ , and  $\Delta X/R$ , (5)–(7), and (10) can be combined and solved for  $k_p$ , with the result given in (11). Note that when using transient virtual impedance for  $Z_{VI}^0$ ,  $R_{VI}^0$  and  $L_{VI}^0$  should be omitted from (11) because the voltage drop associated with  $Z_{VI}^0$  would decay to zero with the high-pass filter's time constant, causing the current to increase beyond  $I_{max}$

$$k_{p,Rvi} = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad (11)$$

where

$$\begin{aligned} a &= (I_{max} - I_{thresh})^2 (1 + (\Delta X/R)^2) \\ b &= 2(I_{max} - I_{thresh}) (R_{VI}^0 + (\Delta X/R) \omega L_{VI}^0) \\ c &= (R_{VI}^0)^2 + (\omega L_{VI}^0)^2 - V_0^2 / I_{max}^2. \end{aligned}$$

### B. Small-Signal Analysis

Small-signal analysis of the inverter-generator microgrid is used to choose  $Z_{VI}^0$  and to investigate the limits on the magnitude and  $X/R$  ratio of  $\Delta Z_{VI}$ . In this paper, the small-signal analysis uses a procedure similar to [5] to write the full differential equations of the inverter, generator, and RL loads and lines. However, instead of linearizing the differential equations by hand, they are simulated in MATLAB Simulink to obtain the steady-state operating point, and MATLAB's "linmod" function is used to linearize the equations around the operating point. In the following small-signal analysis, the operating point is rated load, with a 0.8 power factor linear load.

First, small-signal analysis is used to choose the nominal virtual impedance, i.e.,  $Z_{VI}^0$ . Both the magnitude and  $X/R$  ratio of  $Z_{VI}^0$  impact stability. It has been well documented that a low  $X/R$  ratio causes coupling between the real and reactive power control loops, sometimes causing instability [1], [14], [23]. The root locus plot in Fig. 8 shows the system eigenvalue trajectories when sweeping  $X_{VI}^0/R_{VI}^0$  with a fixed  $|Z_{VI}^0|$ . Two primary eigenvalue pairs of interest are identified:  $\lambda_{15,16}$ , which, based on participation factor analysis ([5], [24]), is associated with the voltage controller states, and  $\lambda_{19,20}$ , which is primarily associated with the generator electromechanical states. Fig. 8 shows that as  $X_{VI}^0/R_{VI}^0$  decreases, damping of the voltage controller mode increases, but damping of electromechanical mode decreases. As  $X_{VI}^0/R_{VI}^0$  drops below 0.5, the electromechanical mode becomes poorly damped.

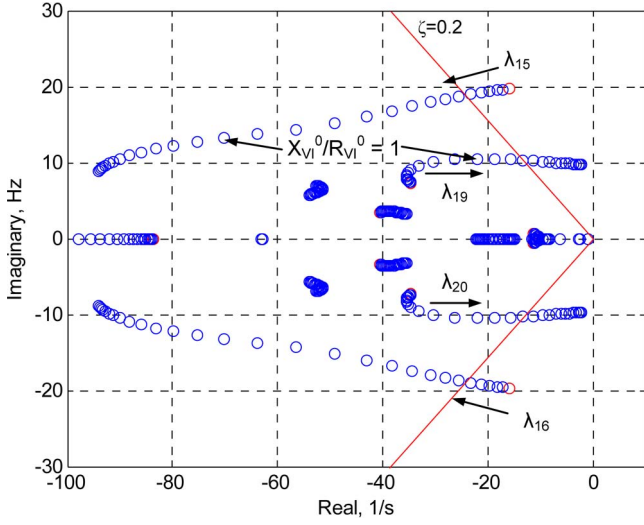


Fig. 8. Root locus for sweeping  $X_{VI}^0/R_{VI}^0$  from 10 to 0.1, with  $|Z_{VI}^0| = 0.1$  pu.

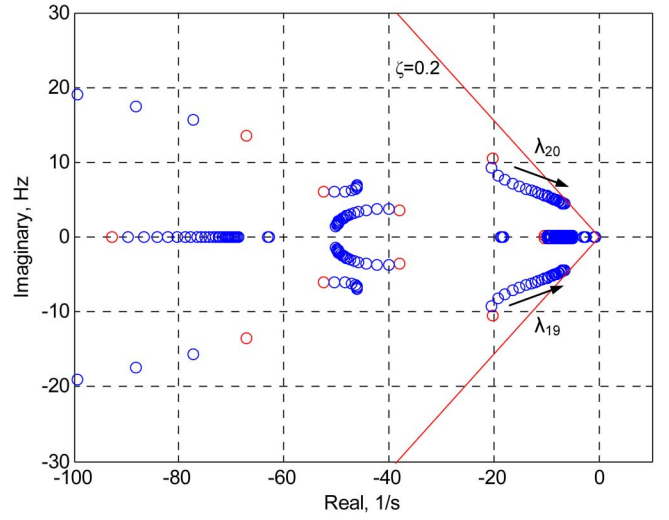


Fig. 10. Root locus for sweeping  $|\Delta Z_{VI}|$  from 0 pu to 0.7 pu with  $\Delta X/R = 1$ ,  $|Z_{VI}^0| = 0.1$  pu, and  $X_{VI}^0/R_{VI}^0 = 1$ .

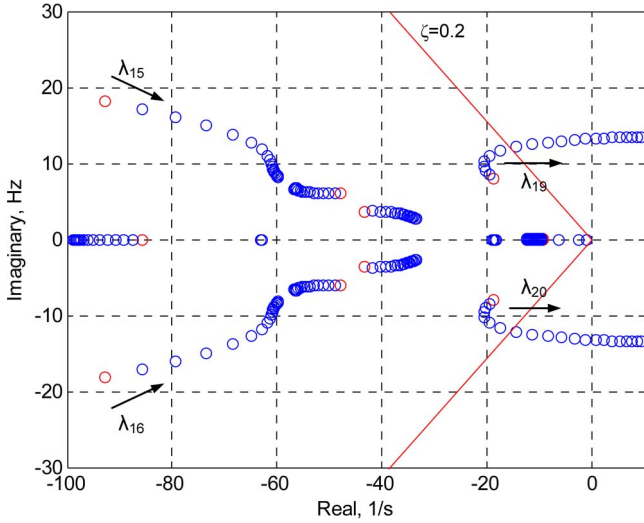


Fig. 9. Root locus for sweeping  $|Z_{VI}^0|$  from 0.15 pu to 0.01 pu with  $X_{VI}^0/R_{VI}^0 = 1$ .

The root locus plot in Fig. 9 shows the eigenvalue trajectories when sweeping  $|Z_{VI}^0|$  with  $X_{VI}^0/R_{VI}^0$  fixed. As  $|Z_{VI}^0|$  is decreased,  $\lambda_{19,20}$  in Fig. 9 become unstable. A compromise between damping of the voltage controller and generator electromechanical mode is chosen at  $|Z_{VI}^0| = 0.1$  pu and  $X_{VI}^0/R_{VI}^0 = 1$  (pointed out in Fig. 8), i.e.,  $R_{VI}^0 = 0.0707$  pu, and  $\omega_0 L_{VI}^0 = 0.0707$  pu.

Next, small-signal analysis is used to give some guidance on the upper stability limits on virtual impedance for current limiting when operating in parallel with the generator. The root locus plot for sweeping  $|\Delta Z_{VI}|$  with a fixed  $\Delta X/R$  and  $Z_{VI}^0$  is shown in Fig. 10. With increasing magnitude of  $Z_{VI}$ , damping of the generator electromechanical mode,  $\lambda_{19,20}$ , decreases. The root locus plot for sweeping  $\Delta X/R$  with a fixed  $|\Delta Z_{VI}|$  is shown in Fig. 11. As  $\Delta X/R$  is increased from resistive to inductive, the damping of the electromechanical mode slightly improves. However, the damping of the high frequency ( $\sim 100$  Hz–150 Hz) voltage controller mode,  $\lambda_{17,18}$ , decreases. For larger

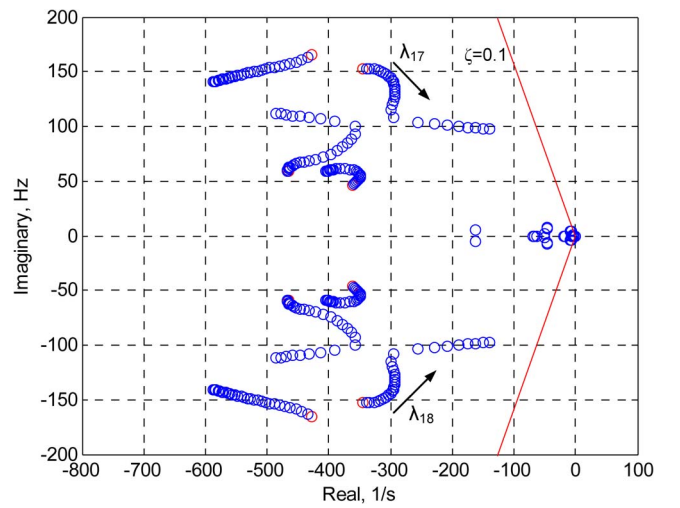


Fig. 11. Root locus for sweeping  $\Delta X/R$  from 1 to 5, with  $|\Delta Z_{VI}| = 0.7$  pu.

$\Delta Z_{VI}$ , such as required when using a smaller  $I_{\max}$  ( $|\Delta Z_{VI}|$  is proportional to  $1/I_{\max}$ ), this high-frequency voltage controller mode can be problematic for inductive  $\Delta X/R$ . Using  $I_{\max} = 2$  pu,  $I_{\text{thresh}} = 1$  pu, and  $\Delta X/R = 5$  in (10) and (11), sufficient damping of both the generator electromechanical mode and the high-frequency voltage controller mode is achieved. The inverter, generator, and system parameters are summarized in Table I.

It is very interesting to note that small-signal analysis shows that too high of a virtual impedance  $X/R$  ratio causes insufficient damping of a voltage controller mode, and it is only very low  $X/R$  ratios ( $< 0.5$ ) that cause instability. This is the case in this experimental low-voltage microgrid with a relatively low  $X/R$  ratio of connecting impedances (see Table I). This indicates that with proper selection of virtual impedance, normal voltage and frequency droop may work acceptably in low-voltage microgrids even with fairly low  $X/R$  ratios. However,  $X/R$  ratio is important, and microgrid designers need to pay attention to coupling impedances.

TABLE I  
INVERTER, GENERATOR, AND SYSTEM PARAMETERS

| Parameter                                                          | Value                             |
|--------------------------------------------------------------------|-----------------------------------|
| Base frequency, $\omega_0$                                         | 60 Hz                             |
| $S_{b\_gen}$                                                       | 15.625 kVA                        |
| $V_{b\_gen}$ , line-neutral                                        | 120 V                             |
| $I_{b\_gen}$                                                       | 43.4 A                            |
| $S_{b\_inv}$                                                       | 11.16 kVA                         |
| $V_{b\_inv}$                                                       | $380/\sqrt{3}$ V                  |
| $I_{b\_inv}$                                                       | 17 A                              |
| Frequency droop slope, $m_p$                                       | $2\pi$ (rad/s)/P <sub>pu</sub>    |
| Voltage droop slope, $m_Q$                                         | $0.05 V_{pu}/Q_{pu}$              |
| P, Q calculation filter cutoff frequency, $\omega_c$               | $10 \times 2\pi$ rad/s            |
| Inv voltage PI proportional gain, $k_{pv}$                         | $0.05 I_{pu}/V_{pu}$              |
| Inv voltage PI integral gain, $k_{iv}$                             | $500 (I_{pu}/V_{pu})/s$           |
| Inv current PI proportional gain, $k_{pc}$                         | $0.5 V_{pu}/I_{pu}$               |
| Inv current PI integral gain, $k_{ic}$                             | $500 (V_{pu}/I_{pu})/s$           |
| Inv current feed-forward gain, H                                   | $0.4 I_{pu}/I_{pu}$               |
| Nominal virtual resistance, $R_{VI}^0$                             | 0.0707 pu                         |
| Nominal virtual inductance, $L_{VI}^0$                             | $0.0707/(2\pi 60)$ pu             |
| Transient VI cutoff frequency, $\omega_{c,hpf}$                    | $2 \times 2\pi$ rad/s             |
| Current limiting max current setting, $I_{max}$                    | 2 pu                              |
| Current limiting threshold, $I_{thresh}$                           | 1 pu                              |
| Virtual resistance current limiting proportional gain, $k_{p,Rvi}$ | 0.098 pu                          |
| Current limiting virtual impedance X/R ratio, $\Delta X/R$         | 5                                 |
| Gen AVR proportional gain, $k_p$                                   | $0.92 V_{pu}/V_{pu}$              |
| Gen AVR integral gain, $k_i$                                       | $23.1 (V_{pu}/V_{pu})/s$          |
| Gen AVR derivative gain, $k_d$                                     | $0.05 (V_{pu}/V_{pu})-s$          |
| Gen Governor $k_p$                                                 | $12 \omega_{pu}/\omega_{pu}$      |
| Gen Governor $k_i$                                                 | $100 (\omega_{pu}/\omega_{pu})/s$ |
| Gen Governor $k_d$                                                 | $0 (\omega_{pu}/\omega_{pu})-s$   |
| Inverter filter inductor impedance                                 | $0.035 + j0.528 \Omega$           |
| Inverter transformer leakage + cable impedance (measured)          | $0.25 + j0.15 \Omega$             |
| Generator cable impedance (measured)                               | $0.04 + j0.01 \Omega$             |

Small-signal analysis was also performed to compare stability with transient virtual impedance versus normal virtual impedance for  $Z_{VI}^0$ . However, there were no significant differences for any of the eigenvalues of interest; thus, no plots are included. This indicates that the distinction between transient virtual impedance versus nonfiltered virtual impedance is not particularly important for system damping, but rather the amount and  $X/R$  ratio of the virtual impedance. Transient virtual impedance is simply used to avoid the steady-state virtual impedance voltage drop caused by positive sequence fundamental current and could be eliminated with the addition of an extra outer voltage loop.

### C. Simulation

A simulation of the virtual impedance current-limiting scheme (controls in Figs. 2 and 7) is shown in Fig. 12 for the same load step as in Fig. 5. The virtual impedance increases when the current magnitude exceeds 1 pu, and the current is reduced from a peak of 1.89 pu without current limiting (see Fig. 5) to 1.25 pu with virtual impedance current limiting. The current limiting comes at the expense of increased voltage dip. The instability caused by current reference saturation limiters is avoided by the virtual impedance current-limiting method by preventing the current references from saturating.

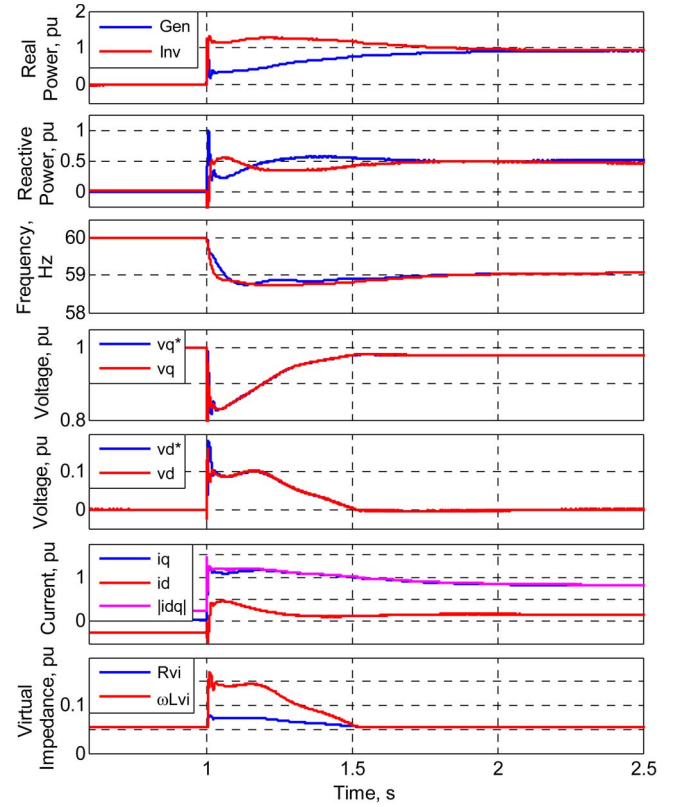


Fig. 12. Simulation of load application with virtual impedance current limiting.

### D. Experimental Results

The proposed controllers have been tested on the experimental microgrid shown in Fig. 3. The generator is a 12.5 kW Marathon Electric Magnapuls 282PSL1704 with a DVR2000E digital voltage regulator. The generator is coupled to an induction motor powered by a variable frequency drive running closed loop speed control to emulate a diesel engine. Inverter control and data acquisition have been implemented in a National Instruments Compact RIO FPGA and real-time processor.

Experimental results for the microgrid in Fig. 3 are shown in Figs. 13–15. All plots show instantaneous, unfiltered quantities. The data acquisition system has problems with picking up switching noise, and is primarily responsible for the significant harmonics present on the power and voltage traces.

Experimental results for the base case (controls in Figs. 2 and 7, with  $k_{p,Rvi} = 0$ ) are shown in Fig. 13. The oscillations of the  $\sim 7$  Hz electromechanical mode are apparent in the power and frequency traces. With simple saturation current limiting, shown in Fig. 14, the  $q$ -axis voltage regulator current limits first, and when the inverter and generator frequency begin to deviate the voltage and current regulators “wind up” and become unstable. Virtual impedance current limiting (controls in Figs. 2 and 7) is shown in Fig. 15. The virtual impedance is inserted when the current magnitude exceeds 1 pu, and reduces the  $q$ -axis voltage reference in order to limit the current. Virtual impedance is seen to be effective for current limiting during overloads caused by poor transient load sharing between GSGFm inverters and synchronous generators.



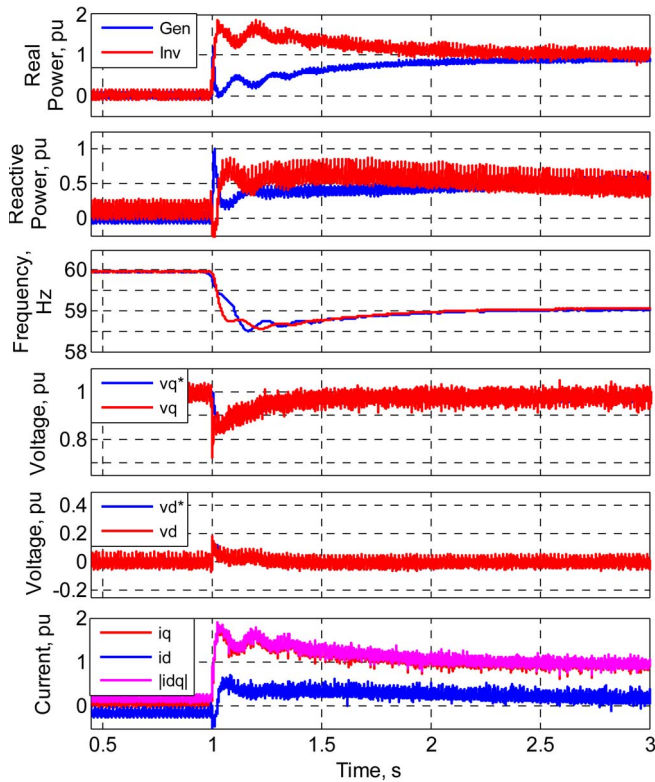


Fig. 13. Experimental results for base case without current limiting.

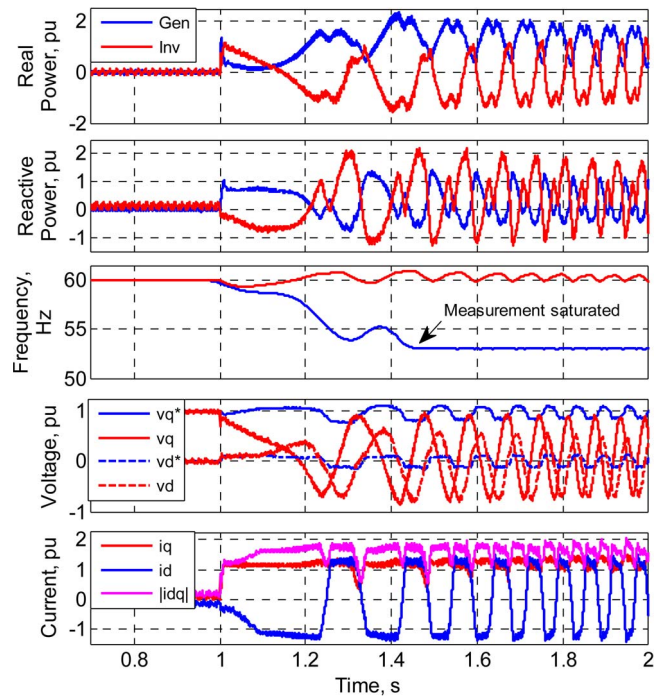


Fig. 14. Experimental results showing instability caused with simple current reference saturation limiting.

## V. CONCLUSION

This paper has analyzed current-limiting controls for voltage control inverters as a solution to poor transient load sharing between inverters and synchronous generators in islanded microgrids. Saturation of the  $d$ - and  $q$ -axis current references can

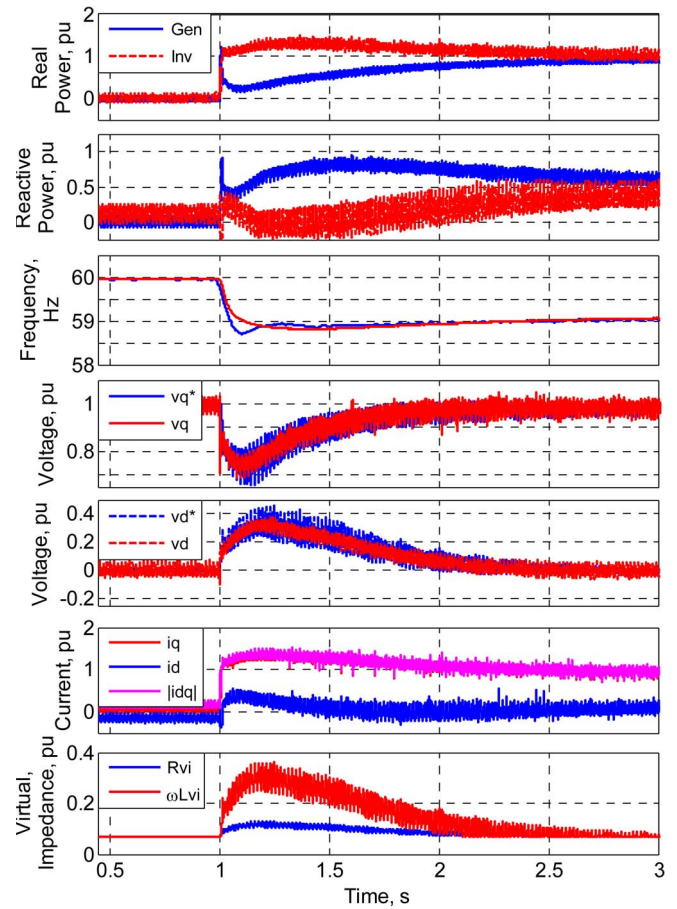


Fig. 15. Experimental results with virtual impedance current limiting showing that the current magnitude is limited and instability avoided.

cause instability when in parallel with synchronous generators by effectively disabling the voltage regulator. Saturation of the current reference ultimately results in divergence of the inverter and generator frequencies, resulting in instability. Virtual impedance current limiting is shown to improve the transient stability by avoiding saturation of the current reference. While virtual impedance current limiting does not inherently prevent the generator and inverter frequencies from diverging, it makes it significantly less likely by modifying the voltage reference instead of saturating the current reference.

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**Andrew D. Paquette** (S'06–M'14) received the B.S. degree in electrical engineering from the University of North Carolina at Charlotte, Charlotte, NC, USA, in 2008, and the M.S. and Ph.D. degrees in electrical engineering from Georgia Institute of Technology, Atlanta, GA, USA, in 2011 and 2014, respectively.

He is currently a Senior Lead Engineer with PC Krause and Associates, West Lafayette, IN, USA, a company that specializes in engineering, modeling, and software solutions for integrated power, thermal, and propulsion systems. His research interests in-

clude control of inverters in microgrids and modeling of integrated aerospace power and thermal systems.



**Deepak M. Divan** (S'78–M'78–SM'91–F'98) is the President and Chief Technology Officer of Varentec, San Jose, CA, USA, a company funded by green-tech venture capital firm Khosla Ventures, which is providing innovative solutions to achieve a smart and dynamically controllable grid. He has over 35 years of experience in industry and academia in the areas of power electronics applied to utility and industrial systems with Varentec being his third entrepreneurial venture. From 2004 to 2011, he served as a Professor of electrical and computer engineering and Founding Director of the Intelligent Power Infrastructure Consortium, Georgia Institute of Technology, Atlanta, GA, USA. Prior to that, he led Soft Switching Technologies as Founder and Chief Executive Officer. He has also been a Professor of electrical engineering with the University of Wisconsin, Madison, WI, USA. He has over 250 papers and 50 issued and pending patents. He combines unique perspectives on the changing landscape on the transmission and distribution grid, and the need for a transition to dynamic grid control, including advanced power electronics solutions. His research interests are in dynamic grid control, sustainable energy, and advanced power electronics.

Dr. Divan was the recipient of the 2006 IEEE Newell Award and was the past President of the IEEE Power Electronics Society.